

Solution Types (with examples)

For the quadratic equation

$$ax^2 + bx + c = 0, \text{ with } a \neq 0 \quad \dots \dots \dots (i)$$

depending on the value of $(b^2 - 4ac)$, there are 3 types of solutions of the quadratic equations:

Case (i) when $(b^2 - 4ac) = 0$

then equation (i) has two identical roots. For example, for the quadratic equation

$$x^2 - 4x + 4 = 0, \quad \dots \dots \dots (ii)$$

$a = 1$, $b = -4$, and $c = 4$, and hence

$$(b^2 - 4ac) = (-4)^2 - 4(1)(4) = 0.$$

Therefore, we expect **two coincident roots**:

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a) = [-(-4) \pm 0] / 2 = 2,$$

are two coincident roots of the given quadratic equation (ii)

Case (ii) when $(b^2 - 4ac) > 0$,

then equation (i) has two **real and distinct** roots. For example, for the quadratic equation

$$x^2 - 6x + 4 = 0, \quad \dots \dots \dots (iii)$$

$a = 1$, $b = -6$, and $c = 4$, and hence

$$(b^2 - 4ac) = (-6)^2 - 4(1)(4) = 36 - 16 = 20.$$

Thus, the roots are

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a) = [-(-6) \pm \sqrt{20}] / 2 = [6 \pm 2\sqrt{5}] / 2 = (3 \pm \sqrt{5})$$

Thus, the **two roots of (iii)** are $x = (3 + \sqrt{5})$, and $x = (3 - \sqrt{5})$.

Case (iii) when $(b^2 - 4ac) < 0$

then equation (i) has two **complex and distinct** roots. For example, for the quadratic equation

$$x^2 - 2x + 4 = 0, \quad \dots \dots \dots (iv)$$

$a = 1$, $b = -2$, and $c = 4$, and hence

$$(b^2 - 4ac) = (-2)^2 - 4(1)(4) = 4 - 16 = -12.$$

Thus, the roots are

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a) = [-(-2) \pm \sqrt{(-12)}] / 2 = [2 \pm 2\sqrt{3}] / 2 = (1 \pm i\sqrt{3})$$

Thus, the **two complex roots** of (iv) are $x = (1 + i\sqrt{3})$, and $x = (1 - i\sqrt{3})$.

Check Your Progress 4

- 1) Arjun is twice as old as Rahim. Five years ago, Arjun's age was three times Rahim's age. Find their present ages.

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- 2) Solve the following equation, which apparently is not linear $(2x+5)/(x+4) = 11/7$

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- 3) Find solutions of the equation $(3x+4y+1)/(2x+3y) = 8/5$

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- 4) Solve each of the following quadratic equations in one variable x:

- i) $x^2 + 6x + 9 = 0$
 ii) $x^2 + 6x + 3 = 0$
 iii) $x^2 + 6x + 12 = 0$

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2.4 LET US SUM UP

In this unit, first we have explained a number of concepts—involved in formulating, defining and solving problems using mathematical tools—including those of symbol, parameter, variable, constant, mathematical expression/formula/well-formed-formula, equation, solution/root of equation, zero of function, polynomial identity, polynomials, and polynomial equations. Then we explained how to apply algebraic operations to polynomials, and how to use polynomial identities in solving specific type of problems. Finally, we discussed how to solve following type of equations: linear equations in one variable single, linear equations in two variables, system of two or more linear equations in two or more variables, and quadratic equations.

2.5 KEY WORDS

- Linguistic Expression** : Linguistic expression is a language, a structure, or a sequence of characters, which is constructed according to its structuring rules--the totality of which is called its *syntax/grammar*.
- Mathematical expression (also called 'formula')** : By this, we mean a *sequence of mathematical terms* (including constants, variables, parameters, relations and operations) formed by using the structuring rules of mathematics.
- Symbol** : A *symbol* is a notation which *represents, or stands for, something else*. For example, the four-letter word *lake* represents some water-body. By '*something else*', we mean that through trying to touch the four-letter word *lake*, our finger does not get wet, but through trying to touch *what is represented* by the four-letter word, finger becomes wet. However, in case we wish to talk about the four-letter word *lake* itself, and not the water-body, then the *symbol* for it is generally written as either of (i) 'lake', (ii) "lake", and (iii) *lake*, or some combination of these, e. g. '*lake*'.
- Variable** : It is a symbol which *can be measured and which may change*. Examples of a variable from the field of economics include price, interest-rate, wages, quantity-bought, and quantity-sold.

2.6 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) (i) and (iv) are M.E. But (ii) is not M. E., because the multiplication symbol '×' must be followed an expression, (iii) is not M. E., because one of the arguments of multiplication symbol '×' is $(8 < y)$, which contains a relational operator, viz. '<', (v) is not M.E., because, it contains two left parentheses, but only one right parenthesis: ***A mathematical expression must contain pairs of matching parentheses.***

Check Your Progress 2

- 1) $(2a - 4b - 5c)^2 = (2a)^2 + (-4b)^2 + (-5c)^2 + 2(2a)(-4b) + 2(2a)(-5c) + 2(-4b)(-5c)$
 $= 4a^2 + 16b^2 + 25c^2 - 16ab - 20ac + 40bc$

- 2) As in Example 2.2.4.1.2, $(625/16) = (25/4)^2$ and $(9/49) = (3/7)^2$. Therefore,
 $(625/16)x^4 - (9/49)y^4 = [(25/4)x^2 + (3/7)y^2] [(25/4)x^2 - (3/7)y^2] \dots (i)$

By further factoring $[(25/4)x^2 - (3/7)y^2] = [(5/2)x + \sqrt{(3/7)y}] [(5/2)x - \sqrt{(3/7)y}] \dots (ii)$

Substituting (ii) in (i), we get

$$(625/16)x^4 - (9/49)y^4 = [(25/4)x^2 + (3/7)y^2] [(5/2)x + \sqrt{(3/7)y}] [(5/2)x - \sqrt{(3/7)y}].$$

- 3) We have $x^2 + 4y^2 + z^2 - 4xy - 4yz + 2xz$
 $= (x)^2 + (-2y)^2 + (z)^2 + 2(x)(-2y) + 2(-2y)(z) + 2(x)(z)$
 $= [x - 2y + z]^2 = [x - 2y + z] [x - 2y + z].$

Check Your Progress 3

- 1) (i) $P(x, y) + Q(x, y) = [4x^2 - 3xy + 5y] + [-4x^2 - 3xy + 5y^2]$

$$= 4x^2 - 3xy + 5y - 4x^2 - 3xy + 5y^2$$

Collecting like terms together

$$= 4x^2 - 4x^2 - 3xy - 3xy + 5y + 5y^2 = -6xy + 5y + 5y^2$$

May be further simplified as

$$= -6xy + 5y(1 + y),$$

Or even, further as

$$= y[-6x + 5(1 + y)]$$

- (ii) $P(x, y) - Q(x, y) = [4x^2 - 3xy + 5y] - [-4x^2 - 3xy + 5y^2]$

Changing the signs of $Q(x, y)$ in the above we get

$$P(x, y) - Q(x, y) = [4x^2 - 3xy + 5y] + [4x^2 + 3xy - 5y^2]$$

Collecting similar terms together,

$$4x^2 + 4x^2 - 3xy + 3xy + 5y - 5y^2$$

$$= 8x^2 + 5y - 5y^2$$

May be further simplified as

$$= 8x^2 + 5y(1 - y).$$

- 2) $(4x^2 - 3xy + 5y) \times (2x^2 - 7y)$

$$= 4x^2 \times (2x^2 - 7y) - 3xy \times (2x^2 - 7y) + 5y \times (2x^2 - 7y)$$

$$= [4x^2 \times 2x^2 + 4x^2 \times (-7y) + [-3xy \times 2x^2 - 3xy \times (-7y)] + [5y \times 2x^2 + 5y \times (-7y)]$$

$$= 8x^4 - 28x^2y - 6x^3y + 21x^2y + 10x^2y - 35y^2$$

Check Your Progress 4

- 1) Let Rahim's current age = x years

Arjun's current age = $2x$ years

5 years earlier Rahim's age = $(x - 5)$ years

5 years earlier Arjun's age = $(2x - 5)$ years

But given Five years ago, Arjun's age was three times

Rahim's age, we get

$$(2x - 5) = 3(x - 5)$$

$$= 3x - 15$$

Transferring all x 's to L.H.S and constants to R.H.S, we get

$$2x - 3x = -15 + 5 = -10$$

$$-x = -10 \text{ implying } x = 10$$

Rahim's current age = 10 years

Arjun's current age = twice Rahim's age = $2(10) = 20$ years.

- 2) Except for the value $x = -7/3$, the equation

$$(2x + 5)/(3x + 7) = 11/16, \dots \dots \dots (i)$$

is equivalent to linear equation

$$(2x + 5) = (11/16)(3x + 7), \text{ or}$$

$$16(2x + 5) = 11(3x + 7) \dots \dots \dots (ii)$$

$$\text{i.e., } 32x + 80 = 33x + 77$$

Shifting terms involving x 's to L.H.S. and the rest to the R.H.S., we get

$$-x = -3$$

Or $x = 3$ is a solution of (ii).

As $3 \neq -7/3$, therefore,

$x = 3$ is a solution of the given equation (i) also.

- 3) Except for $(2x + 3y) = 0$, i. e., for $x = (-3/2)y$, the equation (i) is equivalent to $(3x + 4y + 1) = (8/5)(2x + 3y) \dots \dots \dots (ii)$,

which is a linear equation in two variables x and y , and hence, is expected to have *infinitely* many solutions. From (ii), we get

$$5(3x + 4y + 1) = 8(2x + 3y),$$

equivalently

$$15x + 20y + 5 = 16x + 24y.$$

Shifting y 's to L.H. S. and the rest to R. H.S., we get

$$20y - 24y = 16x - 15x - 5$$

$$-4y = x - 5 \text{ or } y = (-x + 5)/4$$

For the solutions of equation (ii) above, for each value of x , we get a value of y , including the following

$$(x, y) = (1, 1), (2, 3/4), (3, 1/2), \dots$$

For solutions of equation (i), out of these, we reject, if any, those for which $x = (-3/2)y$.

4) Each of the above equation is of the form

$$ax^2 + bx + c = 0,$$

two solutions of which are

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a).$$

For equation (i),

$$a=1, b=6, \text{ and } c=9;$$

Therefore,

$$b^2 - 4ac = (6)^2 - 4(1)(9) = 0$$

Hence the two roots are coincident, and equal to $= -6/2(1) = -3$.

For equation (ii),

$$a=1, b=6, \text{ and } c=3;$$

Therefore,

$$b^2 - 4ac = (6)^2 - 4(1)(3) = 24 > 0.$$

The two roots are real, and are given by $[-6 \pm \sqrt{24}]/2$, i. e

The two roots are given by $[-3 \pm \sqrt{6}]$

For equation (iii),

$$a=1, b=6, \text{ and } c=12;$$

Therefore,

$$b^2 - 4ac = (6)^2 - 4(1)(12) = -12 < 0.$$

The two roots are complex, and are given by $[-6 \pm \sqrt{(-12)}]/2$, i. e

The two roots are given by $[-3 \pm i\sqrt{3}]$.

2.7 EXERCISES

Q1. Factorise $x^2 + 16y^2 + z^2 - 16xy - 16yz + 8xz$.

Ans: We have $x^2 + 4y^2 + z^2 - 4xy - 4yz + 2xz$

$$= (x)^2 + (-2y)^2 + (z)^2 + 2(x)(-2y) + 2(-2y)(z) + 2(x)(z)$$

$$= [x - 2y + z]^2 = [x - 2y + z] [x - 2y + z].$$

Q2. Find the product of $(8x^2 - 6xy + 10y)$ and $(4x^2 - 11y)$

Ans: $(4x^2 - 3xy + 5y) \times (2x^2 - 7y)$

$$= 4x^2 \times (2x^2 - 7y) - 3xy \times (2x^2 - 7y) + 5y \times (2x^2 - 7y)$$

$$= [4x^2 \times 2x^2 + 4x^2 \times (-7y) + [-3xy \times 2x^2 - 3xy \times (-7y)] + [5y \times 2x^2 + 5y \times (-7y)]$$

$$= 8x^4 - 28x^2y - 6x^3y + 21xy^2 + 10x^2y - 35y^2$$

Q3. The digits of a two-digit number differ by 4. If the digits are interchanged, then the original number is one less than double of resulting number. What is the original number?

Ans: A two-digit number, for example, $52 = 5 \times 10 + 2 \times 1$, has 2 at unit's place and 5 at ten's place.

As on interchanging digits, the number decreases, 10's place digit is more than that of unit's place.

Let unit's digit of original number = x

Then 10's digit of original number = $x + 4$

And the *original number* is $(x+4) \times 10 + x = 11x + 40$

On reversing digits, new number's unit's digit = $(x+4)$

new number's 10's digit = x

Therefore, new number = $10x + (x+4) = 11x + 4$.

Original Number = 2 (New number) - 1

Therefore

$$11x + 40 = 2 [11x + 4] - 1 = 22x + 8 - 1 = 22x + 7, \text{ i. e.}$$

$$11x + 40 = 22x + 7$$

For solving the equation, shifting all x 's on one side and constants to the other, we get

$$-11x = 7 - 40 = -33$$

$x = 3$ (unit's digit of original number)

10's digit of the original number = $x + 4 = 7$

Thus, original number = 73

(New number = 37).

Q4. Solve the following system of equations

$$2x + 5y = 16$$

$$3x + 2y = 13$$

Ans:

$$2x + 5y = 16 \quad \dots \dots \dots (i)$$

$$3x + 2y = 13 \quad \dots \dots \dots (ii)$$

In order to get an equation not involving x's, multiply (i) throughout by 3 (*the coefficient of x in (ii)*), and multiply (ii) throughout by 2 (*the coefficient of x in (i)*), we get

$$6x + 15y = 48 \quad \dots \dots \dots (iii)$$

$$6x + 4y = 26 \quad \dots \dots \dots (iv)$$

By (iii) – (iv), we get

$$15y - 4y = 48 - 26 = 22, \text{ i. e.}$$

$$11y = 22,$$

$$y = 2,$$

using $y=2$ in (i) (*we could have used this value of y in (ii), (iii) or (iv)*), we get

$$2x + 5(2) = 16$$

$$2x = 16 - 10 = 6$$

$$x = 3$$

Thus, $x=3$, and $y=2$ is a solution of the system of equations represented by (i) and (ii).

APPENDIX: PROOF TECHNIQUES

In an applied discipline like *Quantitative methods in Economics*, most of the time, only calculations with operations like +, −, ×, ÷, and √ etc. are involved. And, proofs using techniques & rules of inference are rarely required. However, ‘proof’ is the soul of Mathematics, and hence it has to be an essential component of any mathematics curriculum. In this section, we briefly discuss the following well-known techniques of proof in mathematics:

- i) Direct proof and counter examples
- ii) Proof by contradiction and Proof by contrapositive
- iii) Proof by cases
- iv) Existence proof
- v) Proof by mathematical induction
- vi) Proof of equivalences

Generally, for different types of problems, different techniques may be more appropriate, yet, of course, for some problem types, more than one technique may be equally appropriate.

A problem requiring a proof consists explicitly of two parts, viz. (i) what is given as a set of assumed facts/axioms/ postulates etc, all taken together to be called **Hypotheses** and (ii) what is to be proved (to be called **Conclusion**). Apart from these, *implicitly*, a number of definitions, tools/ techniques/ strategies, earlier proved theorems etc. commonly used in such type of proofs, *are also* assumed to be given.

For example, suppose that we are given the problem: *Prove that for all integers m and n , if m and n are odd, then their product ($m \cdot n$) is odd.*

In this problem, *the explicitly given/assumptions* include (i) each of m and n is an integer and (ii) each of m and n is an odd integer. Here, (ii) subsumes (i), i. e. whatever is said under (i) is implied by (ii). *What is to be proved* is: product ' $m \cdot n$ is odd'. *Implicitly*, a number of appropriate definitions, appropriate properties of the involved elements, the valid rules of inference, and proof techniques are also assumed to be given, along with some other assumptions. Appropriate choice of rules of inference and technique(s) play significant role in solving the problem convincingly and efficiently.

Next, we give a sort of working definition of each of the techniques, and then illustrate the defined technique with some suitable example(s).

I) Direct proofs and counter examples

In a direct proof, the proof process begins with *taking Hypotheses* as true. Some required implicit assumptions including other axioms, definitions, previously derived theorems, and rules of inference, are also taken as true. Then using valid rules of inference, properties of the involved elements etc., we try to reach the *conclusion*.

Example: Consider the problem: *Prove that for all integers m and n , if m and n are odd, then their product ($m \cdot n$) is odd.*

Here, definition of *odd integer* is among the implicit assumptions. As m and n are given to be odd integers, by definition, $m = 2 \cdot K_1 + 1$ and $n = 2 \cdot K_2 + 1$, for some integers K_1 and K_2 . Therefore,

$$m \cdot n = (2 \cdot K_1 + 1) \cdot (2 \cdot K_2 + 1) = 4 K_1 K_2 + 2 (K_1 + K_2) + 1$$

(It may be noted that a number of laws about numbers like commutativity of '+' and '.', distributivity of '.' over '+' etc. are among implicit assumptions which are used in the equations above).

By definition of even number, each of $4 K_1 K_2$ and $2 (K_1 + K_2)$ is even and so is their sum $4 K_1 K_2 + 2 (K_1 + K_2)$. Finally, by definition of 'odd'—the remainder is 1 on division by 2--the product $m \cdot n$ is odd.

Refutation through counter examples

*If an intended conclusion is **not** correct*, then the whole proof process need not be gone through. Rather, one *counter-example*, as explained below is sufficient for the purpose.

Example: To prove that the claim that for all integers m and n , if m and n are odd, then their sum t ($m+n$) is odd' is false. The problem is equivalent to showing that for integers m and n , if m and n are odd, then $m+n$ need not be odd.

One of the solution processes may be: Starting the proof as in Example above, let

$m = 2 \cdot K_1 + 1$ and $n = 2 \cdot K_2 + 1$, for some integers K_1 and K_2 . Therefore,

$m + n = (2 \cdot K_1 + 1) + (2 \cdot K_2 + 1) = 2(K_1 + K_2) + 2$, is an even integer. An integer x can be exactly one of (i) x is even and (ii) x is odd. Therefore, x is not odd (as the intended conclusion requires)

The shorter, and possibly better method: for $m =$ (say) 1 and $n =$ (say) 3, $m+n = 1+3 = 4$ is not odd. The one-line proof, of the statement *not* being true is sufficient, through an example. This type of example for refuting a statement is called **refutation through counter-example**.

II) Proof by contradiction and Proof by Contrapositive (also called 'Indirect Proof' methods)

The distinct feature of the technique (along with 'Proof by Contrapositive') is that, ***in the beginning itself, the intended conclusion is assumed to be wrong***. Let H be the hypothesis (may be composed of a number of statements) and C be the intended conclusion. Then, in this technique, $\sim C$, the negation of C , is assumed to be true. And the proof of C from H follows if we are able to derive a contradiction, using in the process, as in Direct proof technique, hypothesis, definitions and earlier proved theorems etc.

A contradiction is a proposition of the form $S \wedge \neg S$ (S may be any statement, which may or may not occur in H or C), where $S \wedge \neg S$ means both S is true and its negation $\neg S$ is also true. **In other words, (S is both true and False simultaneously). This last statement is NOT acceptable in logic, and Mathematics.**

Example: We prove that empty set is a subset of any other set, i.e., to show $\emptyset \subseteq X$, for any set X , where X be an arbitrary set.

Assume that $\emptyset \subseteq X$ is wrong. Then, $\emptyset \not\subseteq X$, which, by definition, is possible if $\exists x \in \emptyset$ such that $x \notin X$. However,

$\exists x \in \emptyset$ is against the definition of $\emptyset$. The situation has arisen out of the assumption ' $\emptyset \subseteq X$ is wrong'. Hence ' $\emptyset \subseteq X$ is wrong' is not possible. Hence, proof by contradiction establishes $\emptyset \subseteq X$, for every set X .